

Texas College of Management & IT

Bachelor of Information Technology (BIT)

Assignment Title:

Fundamentals and Applications of Artificial Intelligence

Submitted By:

Name: _____

LCID: _____

Year/Semester: BIT 2nd Year / 4th Semester

Submitted To:

Instructor Name: Sudarshan Budhathoki

Course: Artificial Intelligence (BIT 353)

Date of Submission: _____

Texas College of Management & IT, Kathmandu

Assignment Details

Title	Assignment
Subject	Artificial Intelligence
Instructor	Sudarshan Budhathoki
Academic Year	Second Year / Fourth Semester

Assignment Guidelines

- **Individual Work:** Complete this assignment on your own. No collaboration is allowed.
- **Language:** Use formal and academic language throughout.
- **Referencing:** Cite all sources in APA 7th edition format.
- **Plagiarism:** Plagiarized submissions will face strict disciplinary action.
- **Format:** Submit as a PDF via Schoolworks Pro Application by the given deadline.
- **Formatting:** Times New Roman, Size 12, 1.5 line spacing, 1-inch margins, page numbers at bottom-right.
- **Diagrams:** Include relevant diagrams, tables, or visual aids with proper labels where applicable.

Task 1: Theoretical Questions (50 Marks)

Answer ALL questions. Each question is based on a chapter of the AI syllabus.

Chapter 1: Introduction to Artificial Intelligence

[10 Marks — Answer both a and b]

Q1 a)

Define Artificial Intelligence (AI). Explain the difference between Narrow AI and General AI with one real-world example of each.

Q1 b)

Identify and briefly explain any FOUR real-world applications of AI across different fields such as healthcare, education, agriculture, or finance.

Chapter 2: Intelligent Agents

[10 Marks — Answer both a and b]

Q2 a)

What is an intelligent agent? Apply the PEAS framework to describe an AI agent used in an online food delivery system.

Q2 b)

Explain the following types of agent programs with a simple example for each: (i) Simple Reflex Agent (ii) Model-Based Agent (iii) Goal-Based Agent (iv) Utility-Based Agent

Chapter 3: Problem Solving

[10 Marks — Answer both a and b]

Q3 a)

What is a Constraint Satisfaction Problem (CSP)? Define its three components — variables, domains, and constraints — and explain with a real-world example such as class timetabling or exam scheduling.

Q3 b)

Explain state space representation in AI. Using a simple navigation problem (e.g., finding a path between cities), define the initial state, goal state, operators, and path cost.

Chapter 4: Searching Techniques

[10 Marks — Answer both a and b]

Q4 a)

Compare Breadth-First Search (BFS) and Depth-First Search (DFS) in terms of: (i) Completeness (ii) Optimality (iii) Time and Space Complexity (iv) Best use case

Q4 b)

Explain the A* search algorithm. How does it differ from Greedy Best-First Search? Why must the heuristic function be admissible for A* to guarantee an optimal solution?

Chapter 5: Knowledge Representation

[10 Marks — Answer both a and b]

Q5 a)

Explain First-Order Predicate Logic (FOPL). Write FOPL statements for: (i) Every person who exercises regularly stays healthy. (ii) Ram is a person. (iii) There exists a doctor who treats all patients.

Q5 b)

Differentiate between Forward Chaining and Backward Chaining. Give a suitable example for each and state when each approach is preferred in a rule-based AI system.

Chapter 6: Machine Learning

[10 Marks — Answer any TWO from a, b, c]

Q6 a)

Explain supervised and unsupervised learning. Give two examples of each and describe one real-world problem where each type is applied.

Q6 b)

Describe the Naive Bayes classifier. How does it use probability to classify data? Explain with a simple example of email spam detection, showing how probabilities are compared.

Q6 c)

Explain the K-Means clustering algorithm step by step. Using the data points {2, 4, 10, 12} with $K=2$, show one full iteration — assign points to clusters and update the centroids.

Chapter 7: Applications of Artificial Intelligence

[10 Marks — Answer any TWO from a, b, c, d]

Q7 a)

Describe the architecture of an Expert System. What are its key components? Explain the expert system development process with a suitable example (e.g., a medical diagnosis system).

Q7 b)

Explain the structure of a Neural Network. Define the role of the input layer, hidden layers, and output layer. How does backpropagation help the network learn from its errors?

Q7 c)

List and explain the key steps in Natural Language Processing (NLP). Apply each step to the sentence: 'The teacher is explaining machine learning to the students.'

Q7 d)

What is Machine Vision? Explain its basic working concepts and give THREE real-world examples where machine vision is currently being used.

Task 2: Practical Tasks (50 Marks)

Complete ALL four practical tasks. Include your Python code and output screenshots in your submission.

Practical 1: Implement BFS and DFS in Python

[20 Marks]

Given the following graph, implement both BFS and DFS in Python to find a path from node 'S' to node 'G':

Graph: $S \rightarrow A, B$ | $A \rightarrow C, D$ | $B \rightarrow D, E$ | $D \rightarrow G$

Task 1a)

Represent the above graph using an adjacency list in Python.

Task 1b)

Write a BFS function that finds and prints the complete path from node 'S' to node 'G'.

Task 1c)

Write a DFS function that finds and prints the complete path from node 'S' to node 'G'.

Task 1d)

Run both programs, show the output, and briefly explain how the traversal order differs between BFS and DFS for this graph.

Practical 2: Decision Tree Classifier on Iris Dataset

[10 Marks]

Use Python and the scikit-learn library to build and evaluate a Decision Tree classifier.

Task 2a)

Load the Iris dataset, split it into 75% training and 25% testing sets, and train a DecisionTreeClassifier with max_depth=4.

Task 2b)

Print the accuracy score and a classification report showing precision, recall, and F1-score for each class.

Task 2c)

Plot a confusion matrix heatmap and explain what it tells you about where the model is making correct and incorrect predictions.

Practical 3: Naive Bayes Text Classifier

Referencing: Cite all sources in APA 7th edition format.[10 Marks]

Build a Naive Bayes text classifier in Python to classify short reviews as Positive or Negative.

Task 3a)

Create a dataset of at least 8 text reviews (4 positive, 4 negative). Use CountVectorizer to convert text into numerical features.

Task 3b)

Train a MultinomialNB model and evaluate it. Print the accuracy score and classification report.

Task 3c)

Test your trained model on two new sentences of your own choice — one positive and one negative — and show the predicted labels.

Practical 4: AI Ethics — Short Case Study

[10 Marks]

Choose ONE of the following AI applications and write a short case study (approximately 300–400 words):

- **Option A:** Facial Recognition Technology in public surveillance
- **Option B:** AI-based Hiring or Recruitment Systems
- **Option C:** Autonomous (Self-Driving) Vehicles

Task 4a)

Briefly describe the chosen AI application and explain how it works.

Task 4b)

Identify and explain TWO ethical concerns associated with this technology (e.g., bias, privacy, accountability).

Task 4c)

Suggest TWO practical measures that could be taken to address these ethical concerns.

Assignment Cover Page

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Marking Scheme

Criteria	Details	Weightage
Content Accuracy	Correctness and relevance of answers.	30%
Clarity & Organization	Logical flow, coherence, and formatting.	20%
Depth of Analysis	Critical thinking and use of examples.	30%
Visual Aids	Effective use of diagrams, tables, code.	10%
Timely Submission	Submitted within the deadline.	10%

Declaration of Originality

I hereby declare that this assignment is entirely my own original work. All sources used have been properly cited in APA 7th edition format. I understand that plagiarism or academic dishonesty is a serious breach of institutional policy and may result in disciplinary action.

Student Name: _____

Student ID: _____

Signature: _____

Date: _____